Sweet Tooth Flight: Olaf's Survival

Student Name: N.D.C.Minsandi

IT Number: IT22324092

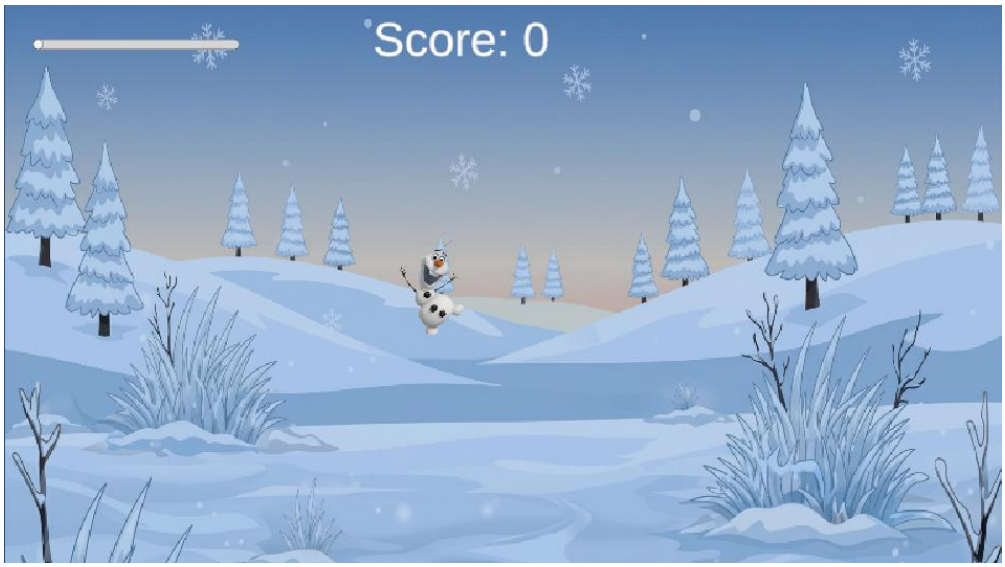
1. Summary of the Game Concept

Sweet Tooth Flight is a 2D arcade-style survival game set in a snowy, winter environment. The player controls Olaf, the snowman from *Frozen*, positioned at the bottom of the screen. The primary objective is to survive as long as possible by dodging deadly fire hazards that fall from the sky. To counteract the melting effects of the fire, the player can catch falling water drops to restore their health. The game tests the player's reflexes and spatial awareness, with the difficulty naturally scaling as the game progresses.

2. Description of How Core Features are Implemented

- **Player Movement:** Movement is handled via a PlayerController C# script attached to the player sprite. It uses Input.GetAxisRaw("Horizontal") combined with a Rigidbody2D component to provide crisp, responsive left and right movement along the X-axis, while the Y-axis remains locked.
- **Hazard and Collectible Spawning:** The falling objects (Fire and Water Drops) are built as Unity Prefabs. A Spawner script is placed at the top of the camera boundary. Using InvokeRepeating or a timer variable, it instantiates these prefabs at randomized X-coordinates so their fall paths are unpredictable.
- **Collision and Health System:** Unity's 2D Physics engine handles all interactions. Both the player and the falling objects utilize Collider2D components set to "Is Trigger". When an object's OnTriggerEnter2D detects a collision with the "Player" tag, it communicates with the GameManager to either subtract health (Fire) or add health (Water Drop), and then destroys the object using Destroy(gameObject).
- **User Interface (UI):** A Unity Canvas overlays the gameplay, featuring a dynamic UI. A Slider component visually represents Olaf's remaining health, and TextMeshPro elements display the current score. The GameManager script ensures these UI elements update in real-time as the player interacts with objects.

3. Screenshots of Gameplay



4. Control Guide

- **Move Left:** W key or Up Arrow key
- **Move Right:** S key or Down Arrow key
- **Objective:** Catch the water drops to heal, and avoid the fire to survive!

5. Description of the Creative Feature

A major creative feature implemented in this game is the **Dynamic Environmental Feedback System**. Instead of relying purely on a UI health bar, the game environment reacts to the player's status. Using Color.Lerp within the GameManager script, the camera's background color smoothly transitions based on the player's health percentage. If Olaf takes fire damage, the snowy background subtly shifts to a warmer, dangerous hue (indicating he is melting). As the player catches water drops and regains health, the environment smoothly cools back down to its original frosty state, providing an immersive, visual warning system.

6. Credits

- **Backgrounds & Icons:** Pinterest
- **Development Tools:** Built using Unity Game Engine.
- **Audio :**Youtube

7. Demo Video link -:
https://drive.google.com/file/d/13UtkD9dIHpj0ikTdEIjod0BnQiJlMqKP/view?usp=sharing